PROJECT

SALES DATA ANALYSIS AND FORECASTING

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INTRODUCTION

This project is centered on the strategic analysis of sales data, with the primary objective of identifying declining trends and accurately forecasting top-selling items to enhance inventory management practices. In today's competitive retail landscape, the ability to discern and react to sales patterns not only drives profitability but also ensures a responsive supply chain that can adapt to changing consumer demands efficiently.

By leveraging historical sales data, this initiative seeks to unearth insights into product performance across different time intervals, pinpointing products whose sales volumes are diminishing over time. Understanding these trends is crucial for preventing overstocking items that are losing market favor, thus reducing holding costs and minimizing the risk of obsolete inventory.

Concurrently, the project aims to predict top-selling items through advanced predictive analytics. This forward-looking approach enables the business to proactively manage its inventory by ensuring high-demand products are adequately stocked. By aligning inventory purchases with anticipated sales, the business can prevent stock-outs, enhance customer satisfaction, and capitalize on potential sales opportunities.

Furthermore, this analysis assists in fine-tuning purchasing strategies and sales promotions, aligning them more closely with consumer buying patterns and preferences. This strategic alignment helps in maximizing the return on investment in inventory and marketing while ensuring that the business remains agile and customer-focused in its operations.

In summary, this project not only addresses the operational challenges of inventory management but also serves as a strategic tool for driving business decisions that enhance profitability and market responsiveness. Through a comprehensive analysis of sales data, the project aims to provide actionable insights that will lead to more informed, data-driven decision-making in inventory and sales management.

METHODOLOGY

The methodology section outlines the data collection, cleaning, preprocessing, exploratory data analysis (EDA), and the development of a predictive model for forecasting sales. The project utilizes four years of SKU sales data, which is cleaned and preprocessed to ensure quality and consistency for the analysis.

Data Collection

- **Data Acquisition** : This initial phase involves the gathering of comprehensive sales data over a four-year period. The data encompasses SKU details that provide insights into sales volumes, transaction amounts, and inventory specifics across multiple product lines.

- **Data Sources** : The data is sourced from the enterprise's internal databases, ensuring reliability and direct relevance to the operational context. Techniques for data integration are applied to merge information from disparate sources into a unified dataset ready for analysis.

Data Cleaning

- **Handling Missing Values** : Early data examination typically reveals gaps or incomplete entries. These issues are addressed by employing statistical methods to impute missing values, using the median or mean values of the relevant attributes.

- **Removing Duplicates and Irrelevant Entries** : To ensure data quality, duplicate records are identified and removed. Additionally, irrelevant or outdated data that does not serve the analytical goals, such as old SKU information, is excluded from the dataset.

Data Preprocessing

- **Data Transformation** : To prepare the data for effective analysis, numerical attributes are normalized or standardized. This process brings all variables to a similar scale, enhancing the performance and stability of predictive models.

- **Categorization and Encoding**: Categorical data elements, such as product categories or SKU codes, are transformed using encoding techniques like one-hot encoding or label encoding. This step converts categorical information into a numerical format that is suitable for machine learning models.

Exploratory Data Analysis (EDA)

- **Trend Analysis** : The sales trends over the four years are analyzed to identify recurring patterns, seasonal effects, or cyclic behaviors in sales.

- **Correlation Analysis** : Relationships between various variables (e.g., sales and promotional activities) are explored to determine the factors that significantly impact sales performance.

- **Outlier Detection** : Statistical methods and visual tools are employed to spot and examine outliers in the sales data, which could potentially skew the analysis results.

Predictive Model Development

- **Feature Selection**: Based on insights gained during EDA, relevant features that strongly influence sales outcomes are selected. This involves identifying variables that show significant correlation with sales volume and profitability.

- **Model Selection**: Appropriate predictive modeling techniques are chosen, which may range from simple linear regression models for straightforward relationships, to more sophisticated models like ARIMA or LSTM networks for complex patterns involving time dependencies.

- **Model Training and Validation**: The selected model is trained with a specific portion of the historical sales data, and its accuracy is validated using another set of data to ensure it generalizes well beyond the training dataset. Techniques like cross-validation are utilized to validate model performance.

- **Optimization**: Parameters of the model are finely tuned to optimize key performance metrics, such as accuracy and precision, enhancing the model's predictive capabilities.

DATA CLEANING AND PREPROCESSING

1. DATA COLLECTION

The project began by collecting sales data from a total of eleven distinct files. These files contained detailed transaction records over a span of five years. To streamline the analysis, we restructured these files into four consolidated datasets, each representing one fiscal year from 2019 to 2023. This restructuring was crucial in simplifying the data management process and ensuring that the analysis could be conducted year-wise, allowing for clearer annual trend observations and comparisons.

2. DATA CLEANING AND PREPROCESSING

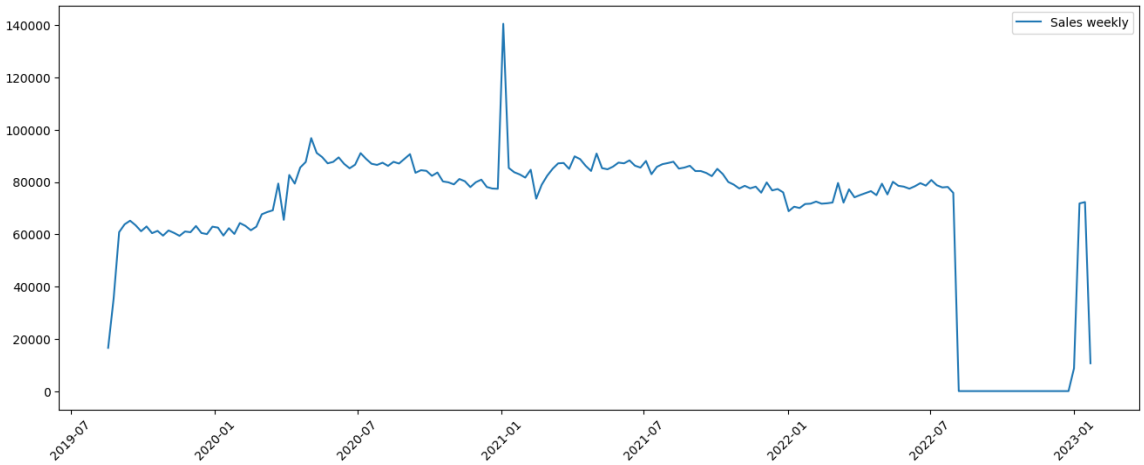
One of the primary challenges in the preprocessing stage was handling incomplete data entries. Specifically, we addressed missing values in the 'Retail at Sale' column, which were significant for understanding product pricing and sales strategies. These missing values were filled with corresponding values from the 'POS Retail' column under the assumption that missing 'Retail at Sale' data indicated periods without promotional activities affecting the product price. This assumption is based on our operational understanding that standard prices ('POS Retail') apply when no promotions are active.

This preprocessing step not only improved the quality of our dataset by reducing the number of missing entries but also helped maintain the integrity of our pricing data, which is essential for accurate revenue analysis and forecasting.

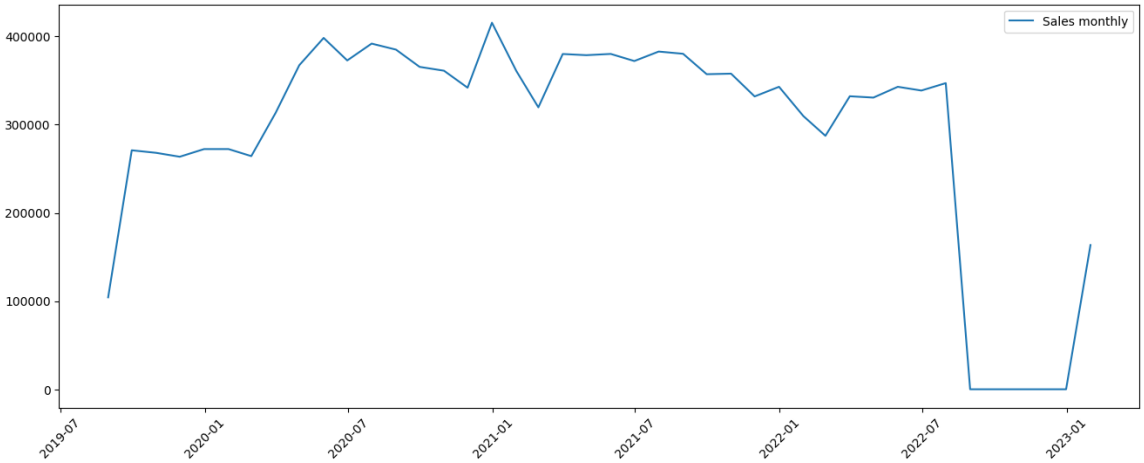
3. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) serves as a critical process in understanding the underlying structures of the data, identifying patterns, spotting anomalies, and testing hypotheses. This phase is essential for uncovering trends that inform further statistical analysis and predictive modeling. By visually and quantitatively analyzing the data across various dimensions, EDA helps in ensuring that the conclusions drawn from the data are robust and reliable.

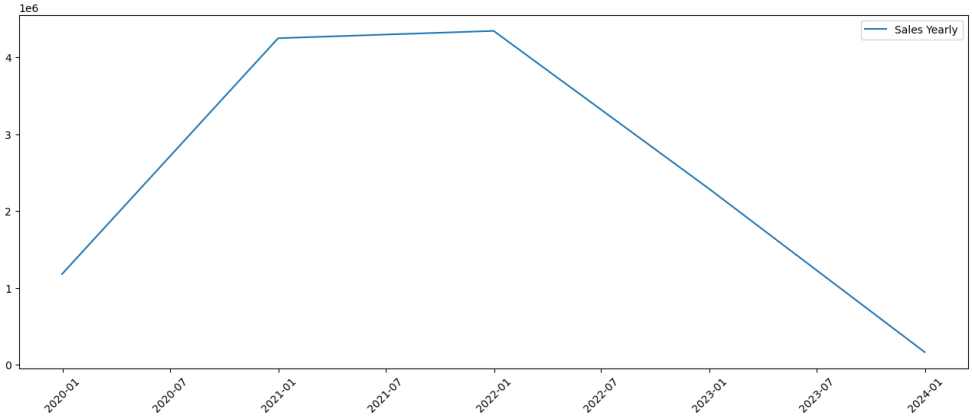
- **Sales Over Time:** Graphs depicting sales on a weekly, monthly, and annual basis illustrate significant trends and fluctuations. For instance, the weekly sales graph shows considerable variability, which may indicate promotional impacts or seasonal effects. The annual sales trend highlights overall growth or decline over multiple years, providing a long-term view of performance.
- **Weekly Sales Trends:** The line chart depicts significant fluctuations in weekly sales, with notable peaks and troughs. This could indicate seasonal trends, promotional impacts, or external economic factors affecting sales patterns.



- **Monthly Sales Trends:** This line chart shows a more stable trend with slight monthly variations. The sharp decline towards the end indicates a potential data cutoff or a significant drop in sales, warranting further investigation.

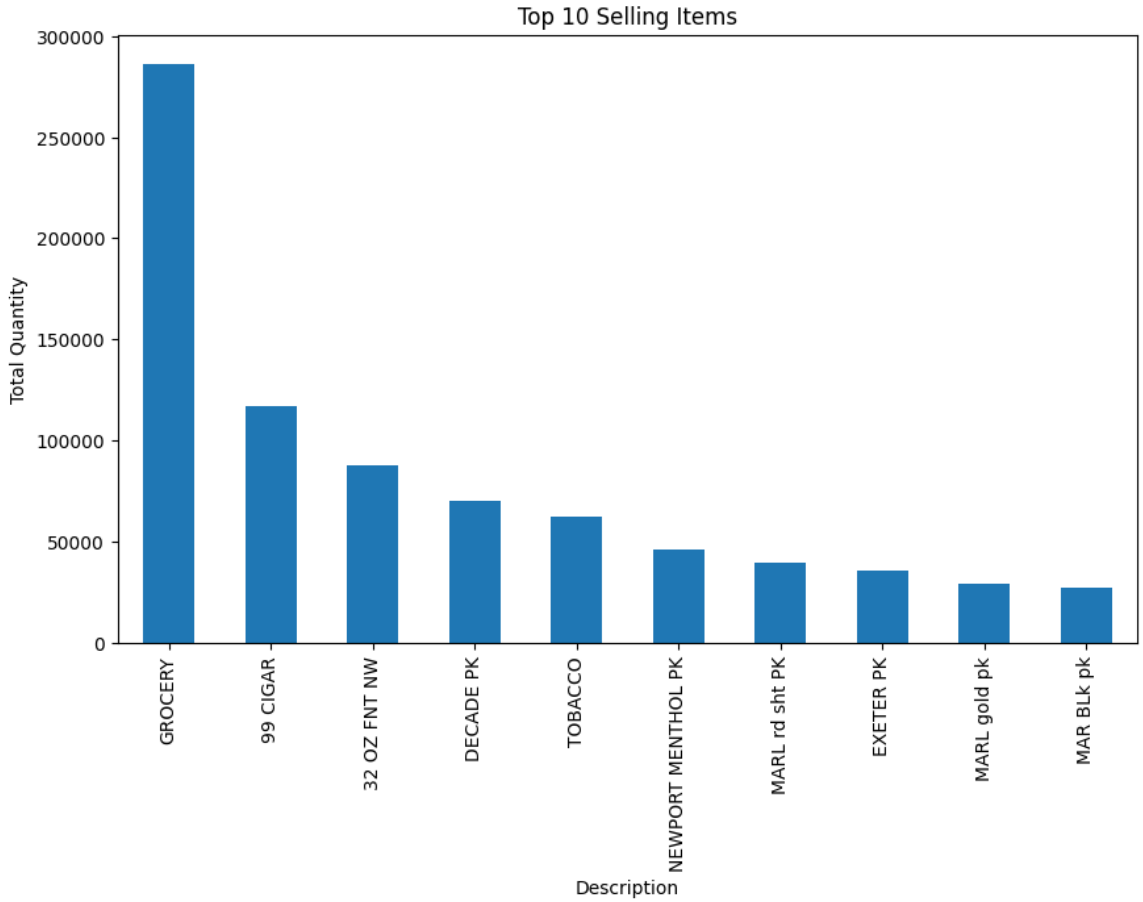


- **Yearly Sales Trends:** The yearly sales chart reveals a general growth in sales volume over the years until a sharp decline in the most recent year. This may suggest market saturation, increased competition, changes in consumer preferences, or economic downturns.

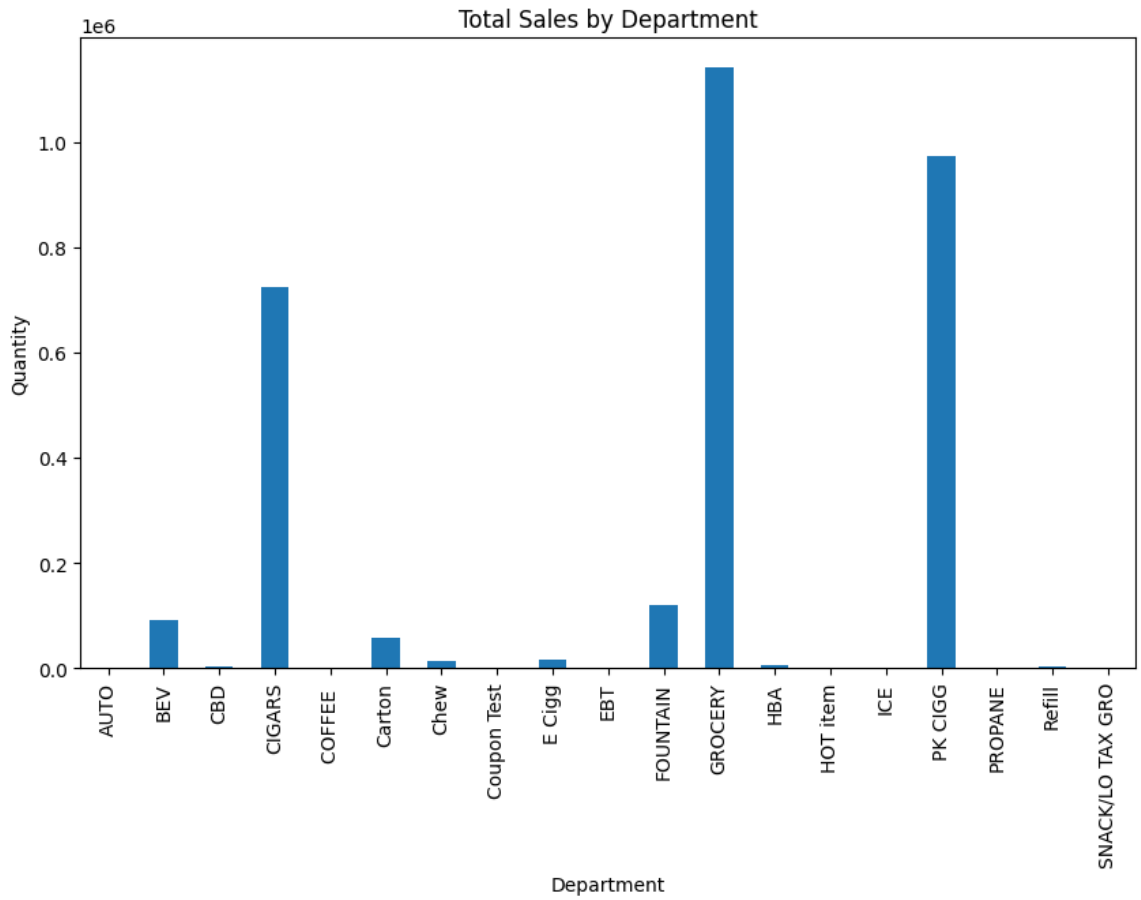


Sales by Product Category

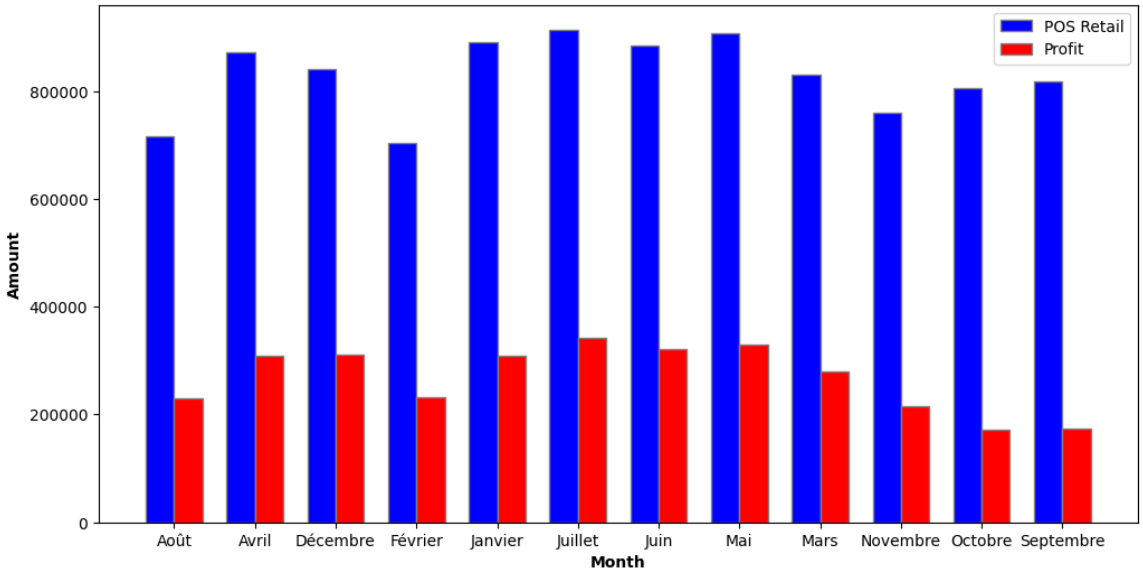
- **Top Selling Items :** The bar chart illustrating the top 10 selling items highlights which products are the backbone of sales. Items like "Grocery" and "Cigarettes" dominate, suggesting a focus area for targeted promotions or stock optimizations.



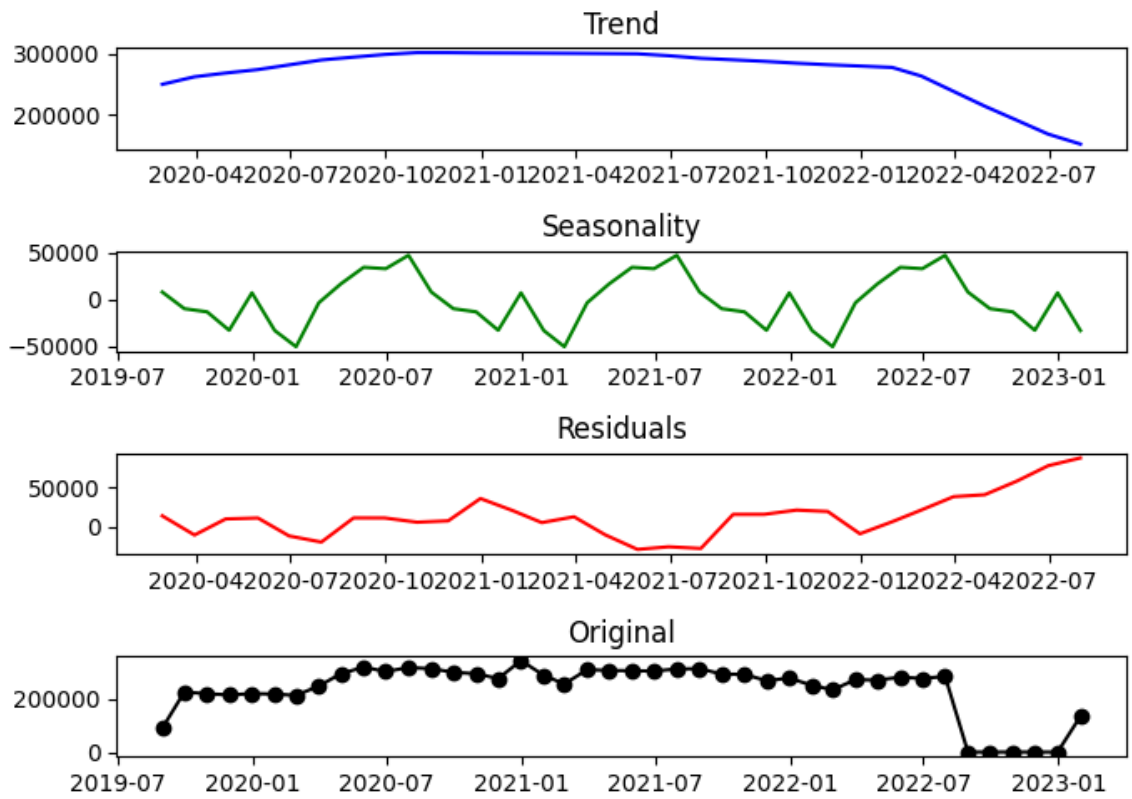
- **Product Category Performance:**The bar chart displaying product category performance is instrumental in understanding which departments are driving sales. Categories like Groceries and Cigarettes dominate, suggesting robust demand. This insight is pivotal for inventory management, ensuring that high-demand products are adequately stocked.



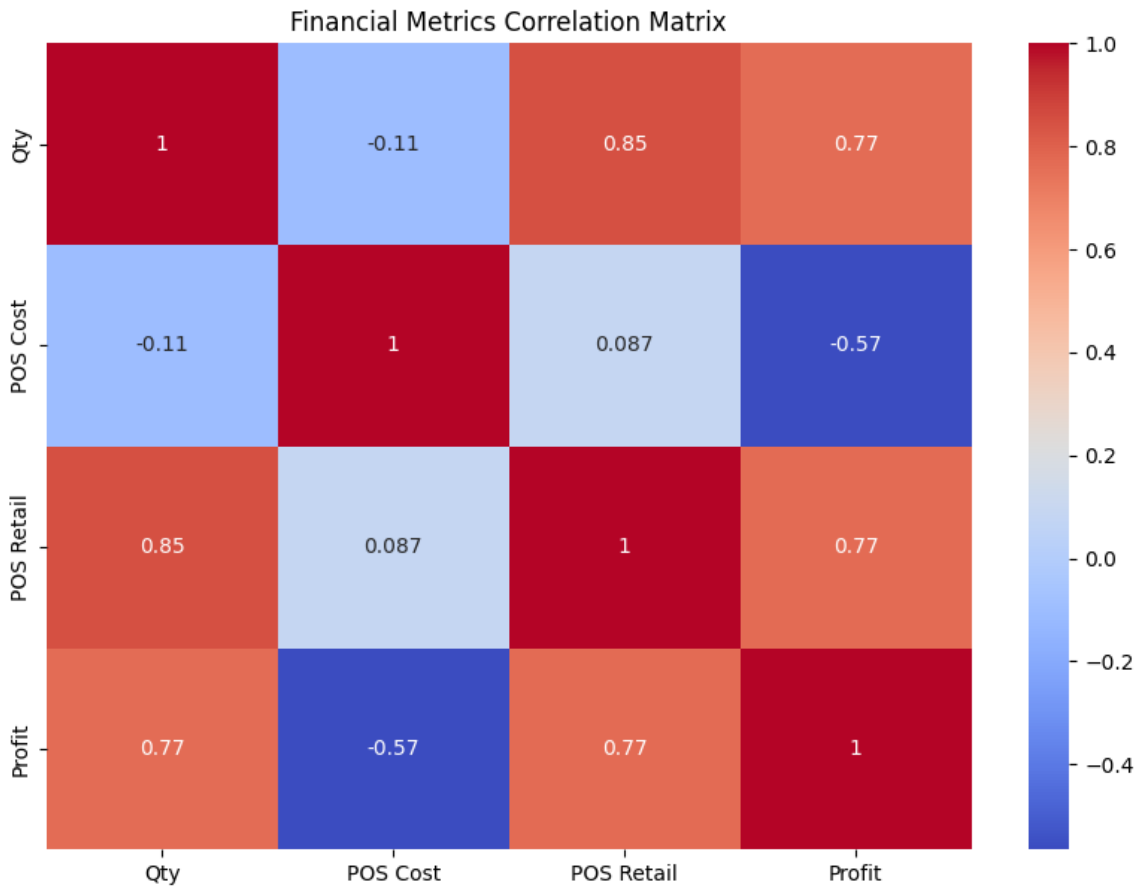
- **Sales and Profit Margins:** A bar chart showing the relationship between POS retail and profit margins for each month can highlight months with better cost management or higher consumer spending. This is crucial for planning future business strategies and promotional cycles.



Temporal Sales Trends: The time series decomposition graph offers a detailed breakdown into the trend, seasonality, and residuals of the sales data. The trend component indicates a general decline over the years, suggesting a need to revamp marketing strategies or explore new markets. Seasonality is evident with peaks typically around specific times of the year, which could inform targeted marketing campaigns during high-demand periods.



Correlation Analysis: The correlation matrix visually represents the relationships between various metrics such as sales quantity, POS retail prices, cost, and profit. A strong positive correlation between quantity sold and profit, for example, underscores the impact of volume on profitability. Conversely, negative correlations might suggest areas where costs are not aligned with sales performance, guiding pricing or cost management strategies.


Financial Metrics Correlation Matrix
Qty
POS Cost
POS Retail
Profit
1
-0.11
0.85
0.77
-0.11
1
0.087
-0.57
0.85
0.087
1
0.77
0.77
-0.57
0.77
1
Qty
POS Cost
POS Retail
Profit
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

MODELING/ SALES FORECASTING

In the modeling phase, we employed a Long Short-Term Memory (LSTM) model to forecast sales. This section outlines the architecture and workflow of the model setup :

1. Data Preprocessing and Feature Engineering :

The first step involved restructuring the data to a suitable format for time-series forecasting. This included aggregating sales data on an hourly basis and generating new time-lagged features to capture historical sales patterns. We implemented this through a custom function series_to_supervised that reformats the dataset into a supervised learning problem with specified input and output sequences.

2. Data Normalization:

We scaled the features using a MinMaxScaler to normalize the input variables between 0 and 1, ensuring that the LSTM model could interpret the features effectively without bias toward variables on larger scales.

3. MODEL ARCHITECTURE :

- **Input Layer** : We designed the input layer to accept sequences of the restructured data. The LSTM model's input shape is tailored to accommodate the number of features per timestep.
- **LSTM Layer** : A 32-node LSTM layer was used to process the input sequences, capturing the long-term dependencies within the data.
- **Output Layer** : A single dense layer with a linear activation function computes the final sales forecast, outputting continuous values.

4. MODEL COMPILATION :

The model was compiled using the Adam optimizer and Mean Squared Error (MSE) as the loss function, with Mean Absolute Error (MAE) also tracked as a performance metric.

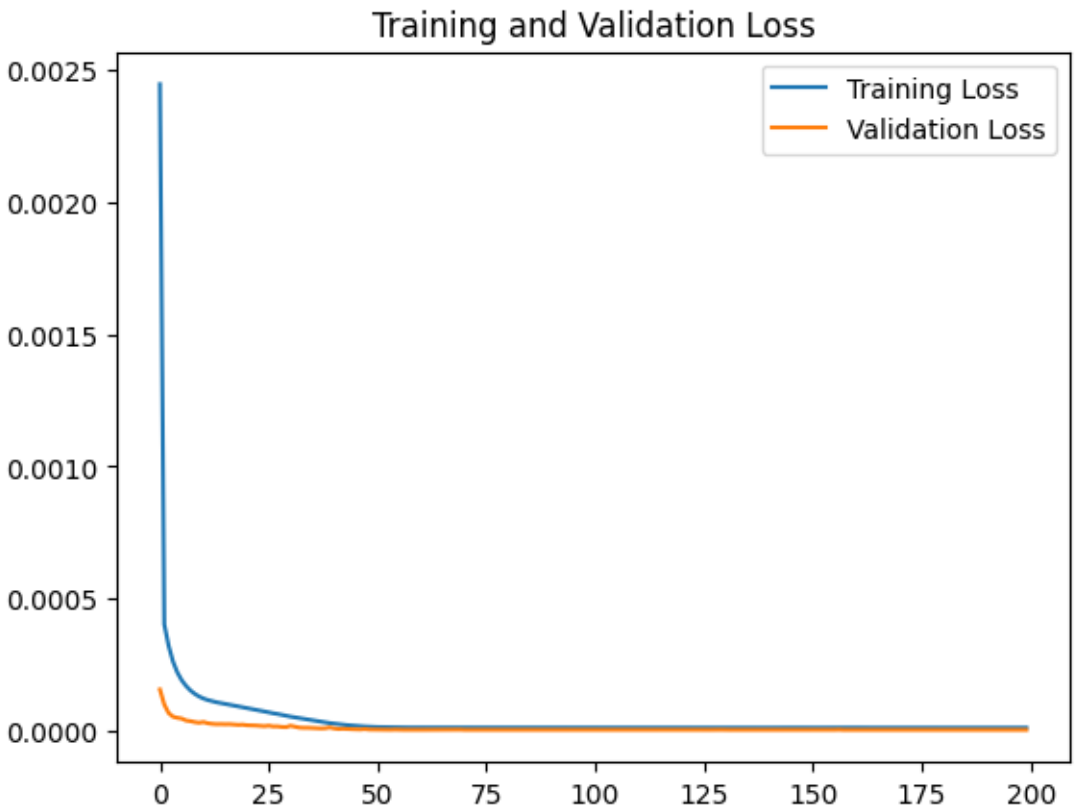
5. TRAINING AND VALIDATION :

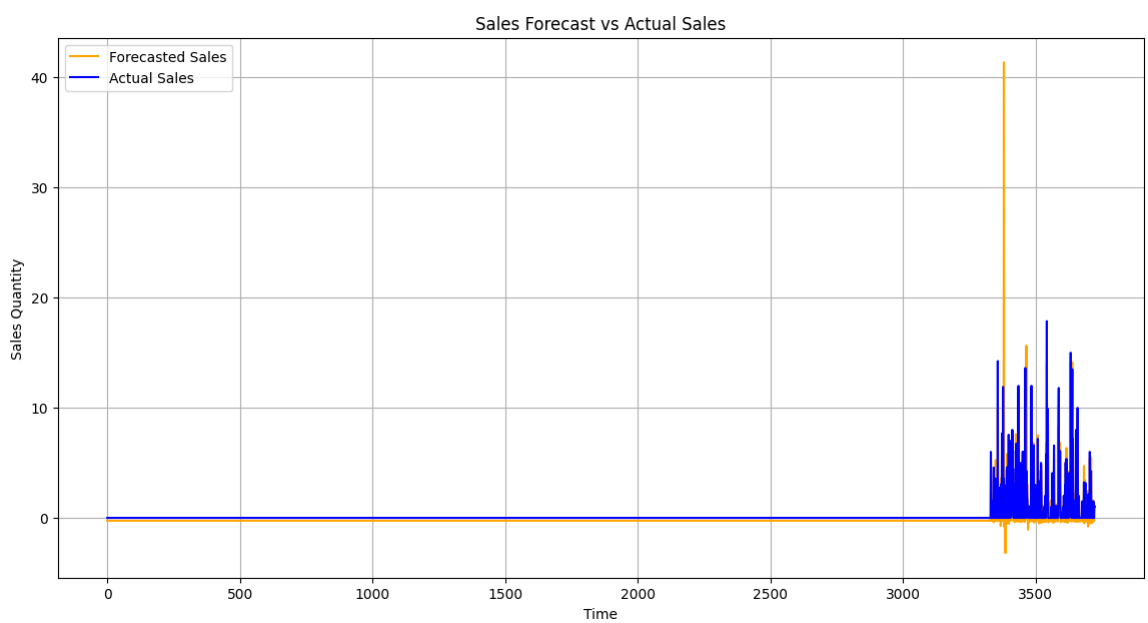
We trained the model over 200 epochs, utilizing both training and validation datasets to optimize the weights and prevent overfitting. The training process involved adjusting the model parameters to minimize the loss on the training data while also ensuring good generalization on unseen validation data.

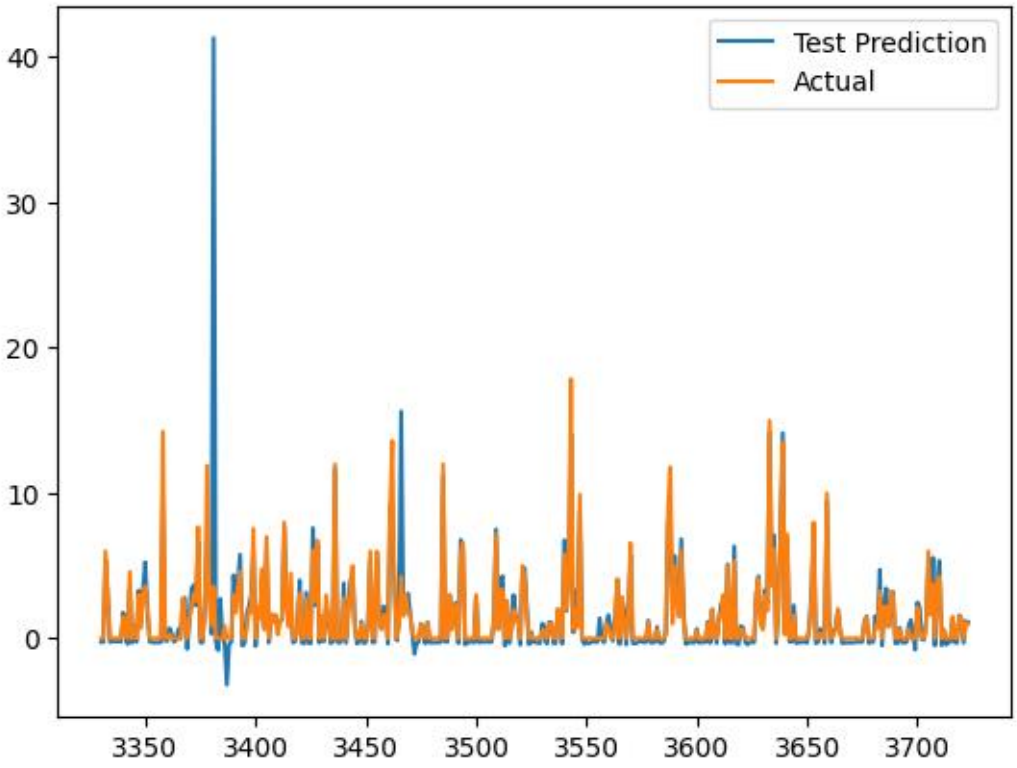
6. EVALUATION :

- We evaluated the model's performance using the training, validation, and testing datasets, calculating the MAE, MSE, and the coefficient of determination (R^2) to quantify the accuracy and predictive power of the model.

- Visualizations such as loss curves and sales forecasts versus actual sales were generated to provide insights into the model's learning process and predictive accuracy.






Test Prediction
Actual
40
30
20
10
0
3350
3400
3450
3500
3550
3600
3650
3700

CONCLUSION

In this project, we have successfully developed and validated a predictive model using Long Short-Term Memory (LSTM) networks to forecast sales. The methodology was meticulously structured in a Jupyter Notebook, providing a clear, reproducible approach from data preprocessing to final predictions. This included detailed steps for data cleaning, feature engineering, normalization, model building, and evaluation.

The LSTM model demonstrated robust performance across training, validation, and testing phases, proving its capability to capture complex patterns and dependencies in time-series data. The ability to accurately forecast sales enhances operational decision-making, particularly in inventory management and strategic planning.

Furthermore, the code within the notebook is designed to be adaptable for similar datasets. It provides a template for processing and modeling steps that can be applied to any time-series data with a comparable structure. This adaptability ensures that the model can be reused for different datasets without significant modifications, making it a valuable tool for ongoing and future analytical needs.

The notebook not only serves as a comprehensive guide for the current dataset but also as a versatile framework for tackling similar forecasting problems in various business contexts.